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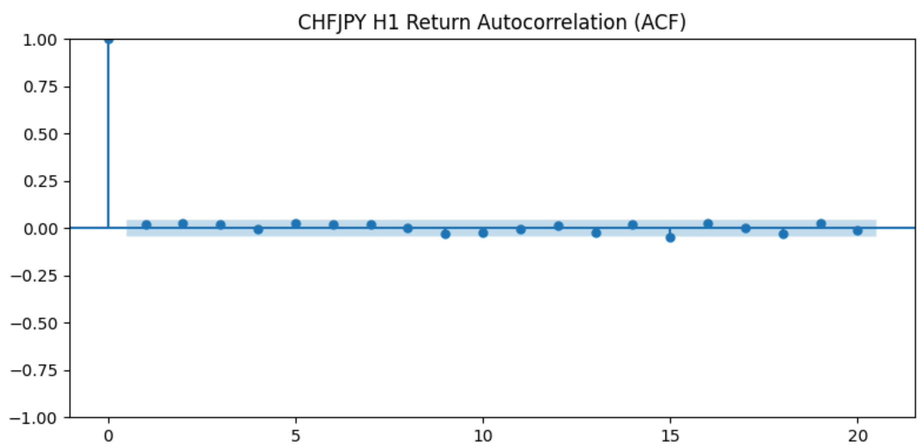
경제학연습: EMA predictive power in FX market(CHFJPY case)

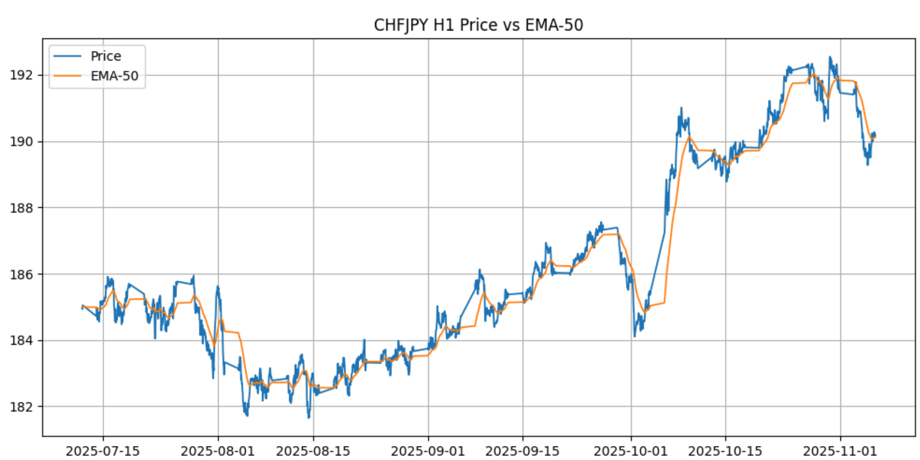
First, after checking in four different timeframes predictive power of EMA50
such as **H1 and H4, where n = 2000 (2.8months and 11months, accordingly)**
D1 and W1, where n = 1000 (2.74 years and 19.7 years, respectively)

(n=1= 1 closed candle per 1H/4H/1D/1W),
Final output interpretation(1) follows as:

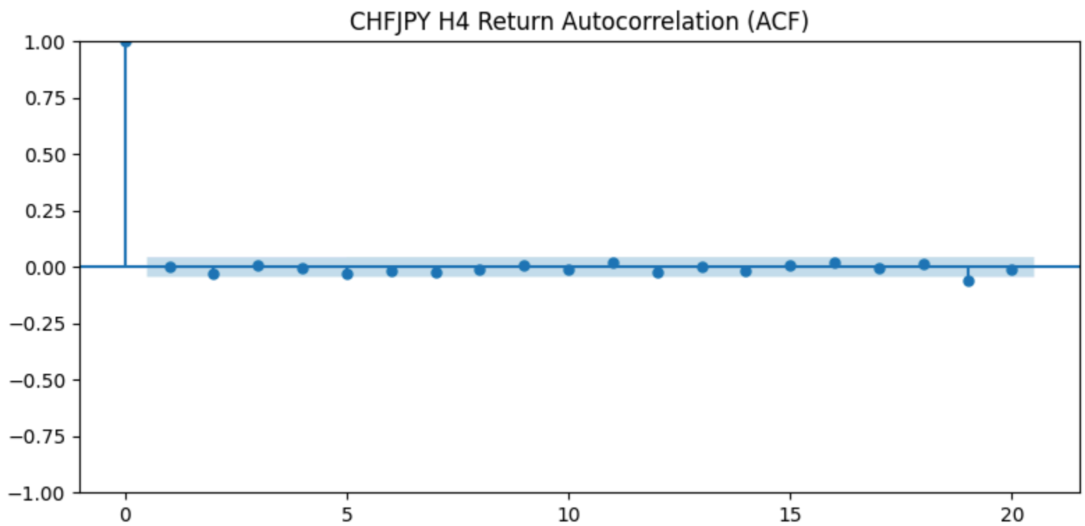
Timeframe	Summary	Market Efficiency
H1	DW>2 => mean reversion. EMA predictive.	Slightly inefficient (reversion pattern)
H4	Stationary, no autocorrelation, DW<2 => positive correlation. EMA predictive.	Inefficient (momentum)

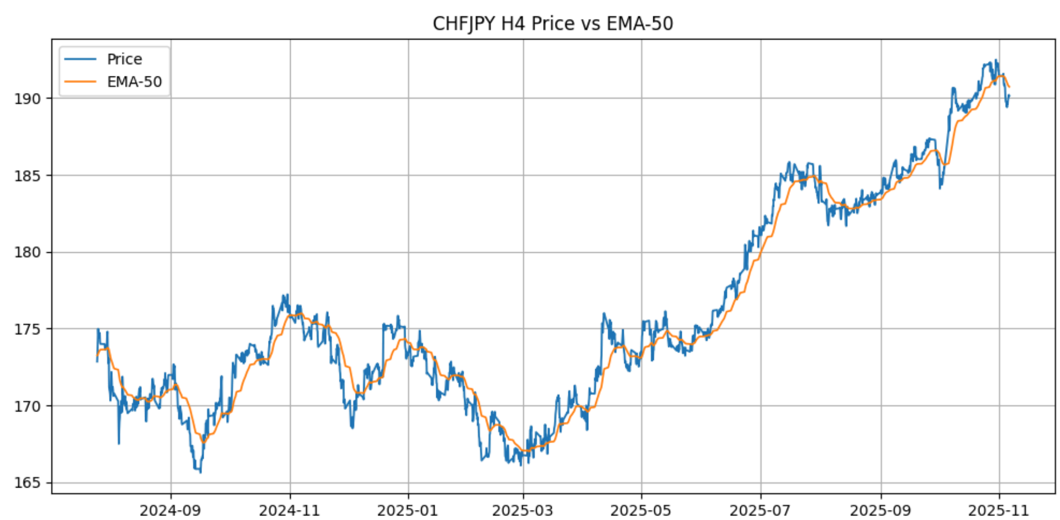
Python Interpretation (1):
=== H1 Timeframe ===
- ADF rejects unit root → returns are stationary → short-term predictability possible.
- No significant autocorrelation → market efficient (no momentum).
- Durbin–Watson=2.00 (>2) → negative correlation (mean reversion).
- EMA signal significantly predicts returns → EMA strategy is statistically justified.





Python Interpretation (2):
=== H4 Timeframe ===
- ADF rejects unit root → returns are stationary → short-term predictability possible.
- No significant autocorrelation → market efficient (no momentum).
- Durbin–Watson=2.00 (<2) → confirms positive serial correlation.
- EMA signal significantly predicts returns → EMA strategy is statistically justified.





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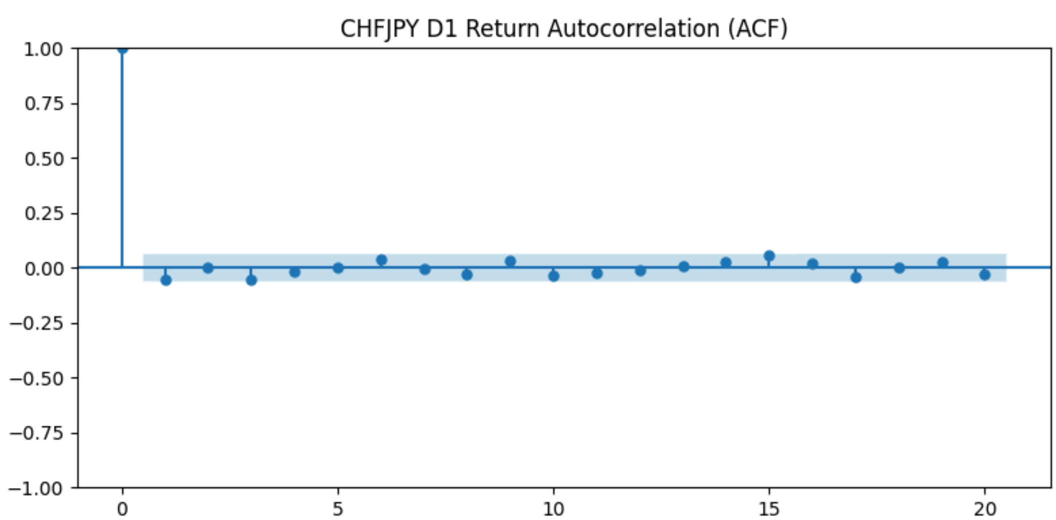
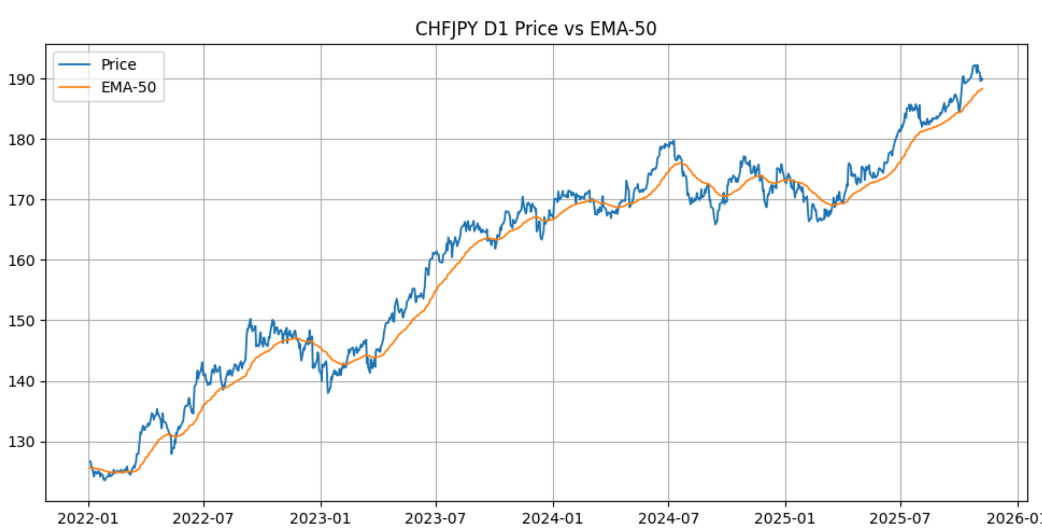
Final output interpretation(2) follows as:

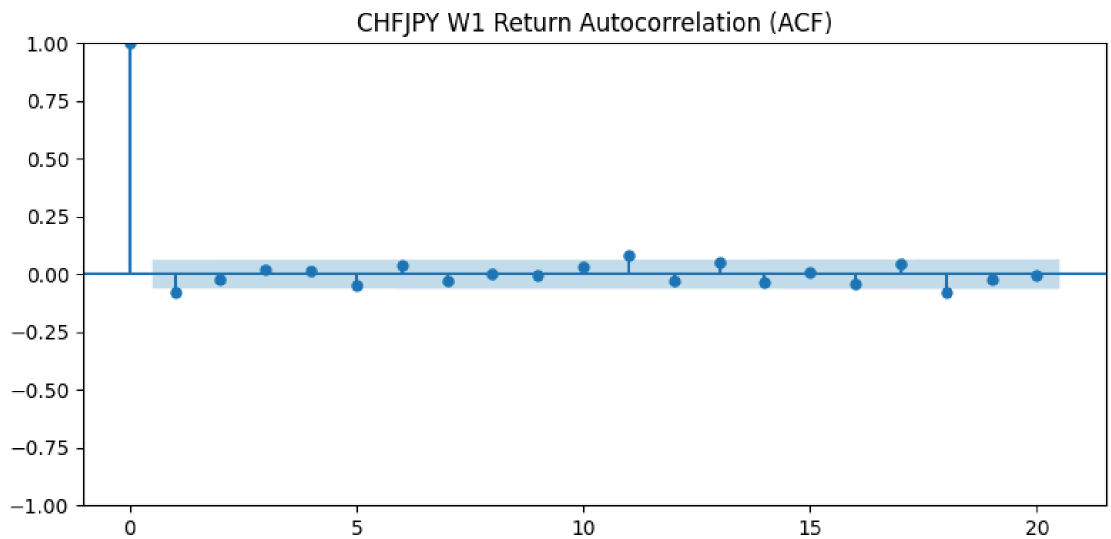
Timeframe	Summary	Market Efficiency
D1	Stationary, no autocorrelation, DW<2 => positive correlation. EMA predictive.	Inefficient (mementum)
W1	Stationary, negative autocorrelation, DW>2 =>mean reversion. EMA predictive	Inefficient(reversion)

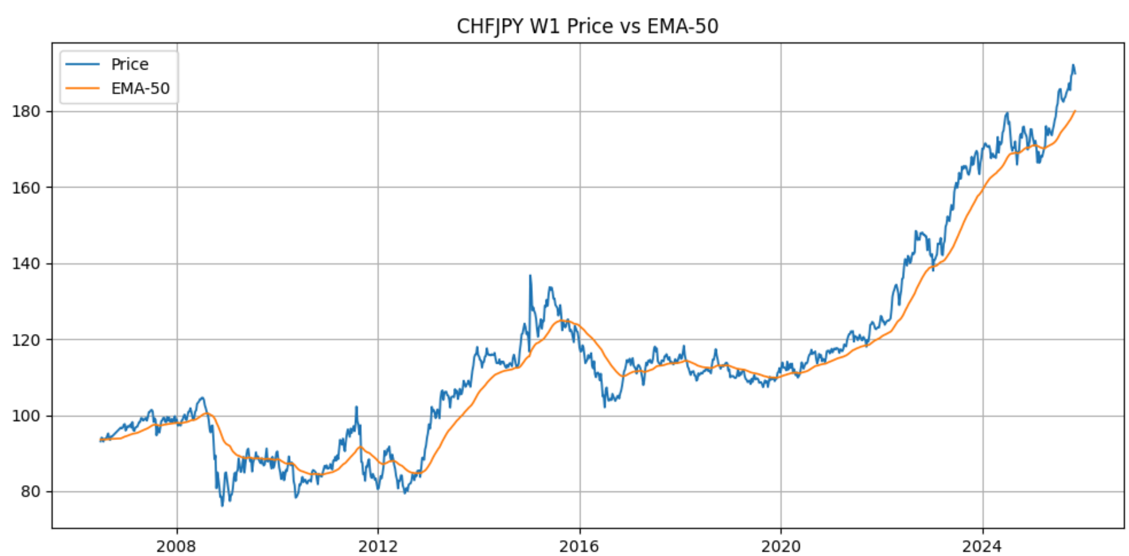
Python Interpretation (1):
=== D1 Timeframe ===
- ADF rejects unit root → returns are stationary → short-term predictability possible.
- No significant autocorrelation → market efficient (no momentum).
- Durbin–Watson=2.00 (<2) → confirms positive serial correlation.
- EMA signal significantly predicts returns → EMA strategy is statistically justified.





Python Interpretation (2):
=== W1 Timeframe ===
- ADF rejects unit root → returns are stationary → short-term predictability possible.
- Negative autocorrelation (φ=-0.08) → mean reversion tendency.
- Durbin–Watson=2.00 (>2) → negative correlation (mean reversion).
- EMA signal significantly predicts returns → EMA strategy is statistically justified.





Overall, the H4 timeframe exhibits the strongest predictive power, primarily due to momentum inefficiency observed within this interval. Statistical evidence indicates a positive serial correlation, suggesting that the EMA-50 aligns well with momentum effects.

However, the predictive strength diminishes as the timeframe increases, partcularly in the D1 and W1 charts, where longer intervals reduce short term responsiveness.

In conclusion, the forecasting power follows the pattern H4 > H1 > D1 > W1 for the CHFJPY currency pair, confirming that the EMA-50 performs with the highest precision on the 4-hour timeframe.
** **Data was run on MetaTrader5, OANDA broker.**